

First Progress Report

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Step we have already achieved:

1. We have created an initial ML workflow diagram. The workflow can change over time if we need it to.
2. We have created the initial preprocessing class where we resize the image and make it greyscale.
3. We have created the initial data split, along with the cross fold validation, with $k = 5$.

Immediate Next Steps:

1. Created the GLCM matrix for feature extraction.
2. Start the initial genetic algorithm optimization code for the LightGBM classifier.
3. Finalize the ML workflow diagram for future users to utilize this model.

Any Changes We To Make:

1. We might need to change the ML workflow depending on what would work best for our project.
2. After we complete the model, we could look at optimizing the code for data loading and preprocessing, in order to take in account memory and runtime limitations.

Overall Plan to complete this project:

Our goal for this project to be complete with the code at least week before the deadline. This way, if we discover any features that we could include into our project or implement any further optimizations we could make, we have enough time to make those changes. This also gives us a good amount of time to work on our presentation and finish the README.md file.

Here is the link to the Github: https://github.com/ayumahapat0/diabetic_retinopathy_prediction/tree/main

Note: We have three branches so far: main, KF, and tanay/data-splitting. We want everyone to work on separate branches so our code doesn't overlap with each other and we work on our own separate features without messing up the other features. In the KF branch, we have an excel sheet tracking our progress.

Here are samples Images (greyscale, original):

